Executive Director's Interview Assessment

From the perspective of an Executive Director (ED) on Nomura's Cash Equities Central Risk Book (CRB) desk, your textbook calculation of the Sharpe Ratio's confidence interval is mechanically correct but displays a dangerous misunderstanding when applied to production risk management.

In a Central Risk Book enviornment handling institutional flow, algorithmic execution residuals, and proprietary inventory positions, evaluating strategy performance using a naive independent and identically distributed (i.i.d.) version of the Sharpe Ratio will lead to structural capital misallocation. The ED will immediately challenge you on two major fronts:

1. **The Compounding Effects of Serial Correlation:** Daily returns of a CRB strategy are virtually never i.i.d. Market impact decay, inventory mean-reversion, and execution scaling strategies introduce severe autocorrelation. If daily returns exhibit positive serial correlation, your sample variance is structurally underestimated, causing your calculated standard errors to be artificially narrow and inflating your backtested Sharpe ratio.
2. **The Scalability Illusion:** You calculated the daily standard error and scaled it by $T = 252$ to represent a 1-year horizon. However, the true metric of concern for a desk head is the **annualized** Sharpe ratio ($\widehat{SR}_{\text{ann}} = \sqrt{252} \cdot \widehat{SR}_{\text{daily}}$). The standard error of the *annualized* metric must scale properly under the delta method, which changes the statistical power calculations for capital allocation.

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1. Deep-Dive Mathematical Derivation

Let us derive the asymptotic distribution of the sample Sharpe Ratio using the **Delta Method**, demonstrating exactly how non-normality enters the variance equation.

Step 1: Definition of Estimators and Point Framework

Let r_t represent the daily excess return of the strategy at time t, sampled from a stationary distribution with mean μ, variance σ^2, skewness $\gamma = \mathbb{E}\left[\left(\frac{r_t-\mu}{\sigma}\right)^3\right]$, and excess kurtosis $\kappa = \mathbb{E}\left[\left(\frac{r_t-\mu}{\sigma}\right)^4\right] - 3$.

The true population daily Sharpe ratio is defined as:

$$SR = \frac{\mu}{\sigma}$$

Given a sample path of T observations, our empirical estimators for the sample mean ($\hat{\mu}$) and sample variance ($\hat{\sigma}^2$) are:

$$\hat{\mu} = \frac{1}{T}\sum_{t=1}^{T} r_t, \quad \hat{\sigma}^2 = \frac{1}{T}\sum_{t=1}^{T}(r_t - \hat{\mu})^2$$

The sample Sharpe ratio estimator is a non-linear function of these two parameters:

$$\widehat{SR} = g(\hat{\mu}, \hat{\sigma}^2) = \frac{\hat{\mu}}{\sqrt{\hat{\sigma}^2}}$$

Step 2: Asymptotic Joint Distribution via Central Limit Theorem

By the Multivariate Central Limit Theorem, the joint sample vector $\boldsymbol{\theta}_T = \begin{bmatrix} \hat{\mu} \\ \hat{\sigma}^2 \end{bmatrix}$ converges asymptotically to a multivariate normal distribution around its true parameters $\boldsymbol{\theta} = \begin{bmatrix} \mu \\ \sigma^2 \end{bmatrix}$:

$$\sqrt{T}\left(\begin{bmatrix} \hat{\mu} \\ \hat{\sigma}^2 \end{bmatrix} - \begin{bmatrix} \mu \\ \sigma^2 \end{bmatrix}\right) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$$

Where $\mathbf{\Sigma} \in \mathbb{R}^{2\times 2}$ is the asymptotic covariance matrix of the estimators:

$$\mathbf{\Sigma} = \begin{bmatrix} \mathrm{Var}(\hat{\mu}) & \mathrm{Cov}(\hat{\mu}, \hat{\sigma}^2) \\ \mathrm{Cov}(\hat{\mu}, \hat{\sigma}^2) & \mathrm{Var}(\hat{\sigma}^2) \end{bmatrix}$$

Let us derive each component of $\mathbf{\Sigma}$ using the central moments of the distribution:

1. **Variance of the Mean (Σ_{11}):**

$$\Sigma_{11} = \lim_{T\to\infty} T \cdot \mathrm{Var}(\hat{\mu}) = \mathrm{Var}(r_t) = \sigma^2$$

2. **Covariance between Mean and Variance ($\Sigma_{12} = \Sigma_{21}$):**

$$\Sigma_{12} = \mathbb{E}[(r_t - \mu)(r_t - \mu)^2] = \mathbb{E}[(r_t - \mu)^3] = \gamma\sigma^3$$

3. **Variance of the Variance (Σ_{22}):**

$$\Sigma_{22} = \mathbb{E}\left[\left((r_t - \mu)^2 - \sigma^2\right)^2\right] = \mathbb{E}[(r_t - \mu)^4] - \sigma^4 = (\kappa + 3)\sigma^4 - \sigma^4 = (\kappa + 2)\sigma^4$$

Assembling our exact covariance structure matrix:

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma^2 & \gamma\sigma^3 \\ \gamma\sigma^3 & (\kappa + 2)\sigma^4 \end{bmatrix}$$

Step 3: Application of the Delta Method

The Delta Method states that for a continuously differentiable function $g(\boldsymbol{\theta})$, the asymptotic variance of $g(\boldsymbol{\theta}_T)$ is given by:

$$\mathrm{Var}\left(g(\boldsymbol{\theta}_T)\right) \approx \frac{1}{T} \boldsymbol{\nabla} g(\boldsymbol{\theta})^\top \mathbf{\Sigma} \boldsymbol{\nabla} g(\boldsymbol{\theta})$$

Let us calculate the gradient vector $\boldsymbol{\nabla} g(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\partial g}{\partial \mu} \\ \frac{\partial g}{\partial \sigma^2} \end{bmatrix}$:

$$\frac{\partial g}{\partial \mu} = \frac{1}{\sqrt{\sigma^2}} = \frac{1}{\sigma}$$

$$\frac{\partial g}{\partial \sigma^2} = \mu\left(-\frac{1}{2}(\sigma^2)^{-3/2}\right) = -\frac{\mu}{2\sigma^3} = -\frac{SR}{2\sigma^2}$$

Now, let us substitute the gradient components and covariance matrix into the quadratic form:

$$\boldsymbol{\nabla} g(\boldsymbol{\theta})^\top \mathbf{\Sigma} = \begin{bmatrix} \frac{1}{\sigma} & -\frac{SR}{2\sigma^2} \end{bmatrix} \begin{bmatrix} \sigma^2 & \gamma\sigma^3 \\ \gamma\sigma^3 & (\kappa + 2)\sigma^4 \end{bmatrix} = \begin{bmatrix} \sigma - \frac{1}{2}\gamma\sigma SR & \gamma\sigma^2 - \frac{1}{2}(\kappa + 2)\sigma^2 SR \end{bmatrix}$$

Next, compute the full scalar inner product $\boldsymbol{\nabla} g(\boldsymbol{\theta})^\top \boldsymbol{\Sigma} \boldsymbol{\nabla} g(\boldsymbol{\theta})$:

$$\text{Var}(\widehat{SR}) \cdot T = \left(\sigma - \frac{1}{2}\gamma\sigma SR\right)\left(\frac{1}{\sigma}\right) + \left(\gamma\sigma^2 - \frac{1}{2}(\kappa+2)\sigma^2 SR\right)\left(-\frac{SR}{2\sigma^2}\right)$$

$$\text{Var}(\widehat{SR}) \cdot T = 1 - \frac{1}{2}\gamma SR - \frac{1}{2}\gamma SR + \frac{1}{4}(\kappa+2)SR^2$$

$$\text{Var}(\widehat{SR}) \cdot T = 1 - \gamma SR + \frac{1}{4}\kappa SR^2 + \frac{2}{4}SR^2$$

$$\text{Var}(\widehat{SR}) \cdot T = 1 + \frac{1}{2}SR^2 - \gamma SR + \frac{\kappa}{4}SR^2$$

Taking the square root and dividing by T isolates the exact asymptotic Standard Error formula under non-normality (Mertens/Lo correction):

$$\boxed{\text{SE}(\widehat{SR}) = \sqrt{\frac{1}{T}\left(1 + \frac{1}{2}SR^2 - \gamma SR + \frac{\kappa}{4}SR^2\right)}}$$

Note: In the simplified version often cited in interview templates, the final kurtosis term is listed as $\frac{\kappa}{4}$ instead of $\frac{\kappa}{4}SR^2$. In practice, the full derivation shows that the kurtosis penalty scales with the magnitude of the Sharpe Ratio squared.

Step 4: Step-by-Step Power Derivation for the Horizon Test

To evaluate the statistical power of testing $H_0 : SR = 0$ against an alternative hypothesis $H_1 : SR = SR_1 > 0$, we establish our decision boundary under a significance level $\alpha = 0.05$.

Under H_0, $SR = 0$. The standard error simplifies significantly to:

$$\text{SE}_0 = \sqrt{\frac{1}{T}(1 + 0 - 0 + 0)} = \frac{1}{\sqrt{T}}$$

For a one-tailed test, we reject H_0 if our test statistic exceeds the critical value:

$$\widehat{SR} > z_{1-\alpha} \cdot \text{SE}_0 = \frac{z_{1-\alpha}}{\sqrt{T}}$$

Statistical power ($1 - \beta$) is defined as the probability of correctly rejecting the null hypothesis given that the true parameter is SR_1. We evaluate this probability under the distribution defined by H_1:

$$\text{Power} = \mathbb{P}\left(\widehat{SR} > \frac{z_{1-\alpha}}{\sqrt{T}} \,\middle|\, \mathcal{N}\left(SR_1, \text{SE}_1^2\right)\right)$$

Standardizing the distribution under H_1:

$$\text{Power} = \mathbb{P}\left(\frac{\widehat{SR} - SR_1}{\text{SE}_1} > \frac{\frac{z_{1-\alpha}}{\sqrt{T}} - SR_1}{\text{SE}_1}\right) = \Phi\left(\frac{SR_1 - \frac{z_{1-\alpha}}{\sqrt{T}}}{\text{SE}_1}\right)$$

Substituting the i.i.d. normal formulation for $\text{SE}_1 = \sqrt{\frac{1+\frac{1}{2}SR_1^2}{T}}$:

$$\boxed{\text{Power} = \Phi\left(\frac{SR_1\sqrt{T} - z_{1-\alpha}}{\sqrt{1 + \frac{1}{2}SR_1^2}}\right)}$$

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2. Application of Theory & Code Portions

The Risks of Naive Estimation

When executing automated market-making or arbitrage frameworks within a Central Risk Book, a sample Sharpe ratio derived from historical trading data can look highly attractive due to short-term data characteristics.

To prevent allocating capital to strategies that look profitable purely due to random noise, a production-grade validation platform must implement three features:

1. **Higher-Moment Estimations:** Explicitly compute the empirical skewness and excess kurtosis to adjust the error bounds.
2. **Autocorrelation Correction (Newey-West Expansion):** Compute the spectral density at zero frequency to account for serial dependence in daily returns.
3. **Annualized Variance Mapping:** Map the daily confidence bounds directly into an annualized framework using the appropriate delta scaling factors.

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3. Production-Grade Institutional Implementation

The following implementation provides an institutional performance validation engine that processes strategy return streams, calculates higher-order moments, corrects for autocorrelation, and outputs robust, non-normal confidence intervals.

```
# ==============================================================================
# crb_sharpe_inference_engine.py
#
# Production-Grade Performance Inference Engine for Cash Equities CRB.
# Implements Lo-Mertens Higher-Moment Adjustment and Newey-West Correction.
# ==============================================================================

from __future__ import annotations

import logging
from dataclasses import dataclass
from typing import Final

import numpy as np
import scipy.stats as stats

# ----------------------------------------------------------------------------
# Setup & Constants
# ----------------------------------------------------------------------------
LOG_FMT: Final[str] = "%(asctime)s | %(levelname)-8s | %(name)s | %(message)s"
```

```
logging.basicConfig(level=logging.INFO, format=LOG_FMT)
logger = logging.getLogger(__name__)


@dataclass(frozen=True, slots=True)
class SharpeInferenceMetrics:
    """Container holding statistically adjusted Sharpe metrics and confidence
bounds."""
    daily_sharpe: float
    annualized_sharpe: float
    standard_error_iid_normal: float
    standard_error_adjusted: float
    confidence_interval_lower: float
    confidence_interval_upper: float
    p_value_zero_null: float


class CRBSharpeInferenceEngine:
    """Inference engine to compute heteroskedasticity and autocorrelation robust
Sharpe metrics."""

    def __init__(self, annualized_factor: int = 252, confidence_level: float =
0.95) -> None:
        self._factor = annualized_factor
        self._alpha_level = 1.0 - confidence_level

    def analyze_returns(self, daily_returns: np.ndarray, lag_hac: int = 3) ->
SharpeInferenceMetrics:
        """Performs structural inference on an array of daily excess strategy
returns.

        Args:
            daily_returns: (T,) Array of daily excess returns.
            lag_hac: Number of autoregressive lags to use for the Newey-West
adjustment.
        """
        T = daily_returns.shape[0]
        if T < 10:
            raise ValueError("Insufficient sample size to compute robust higher-
order moments.")

        mean_r = float(np.mean(daily_returns))
        std_r = float(np.std(daily_returns, ddof=1))

        # Calculate the base point estimate
        daily_sr = mean_r / std_r
        ann_sr = daily_sr * np.sqrt(self._factor)

        # Compute higher-order sample moments
        skew = float(stats.skew(daily_returns, bias=False))
        kurt = float(stats.kurtosis(daily_returns, bias=False)) # Excess kurtosis

        logger.info("Sample summary: T=%d, Skew=%.3f, Excess Kurtosis=%.3f", T,
skew, kurt)
```

```
        # 1. Base i.i.d. Normal Standard Error calculation
        se_iid_normal = np.sqrt((1.0 + 0.5 * (daily_sr ** 2)) / T)

        # 2. Non-Normal Adjustment (Lo-Mertens Expansion)
        # SE^2 = (1 + 0.5*SR^2 - Skew*SR + Kurt/4*SR^2) / T
        variance_adjusted_core = 1.0 + 0.5 * (daily_sr ** 2) - (skew * daily_sr) +
(kurt / 4.0 * (daily_sr ** 2))

        # 3. Apply Newey-West Autocorrelation Correction if serial correlation is
present
        # Adjust variance based on the covariance of lagged terms
        hac_adjustment = 0.0
        demeaned_returns = daily_returns - mean_r
        for lag in range(1, lag_hac + 1):
            weight = 1.0 - (lag / (lag_hac + 1))
            # Calculate autocorrelation covariance at this lag step
            covariance = float(np.mean(demeaned_returns[lag:] *
demeaned_returns[:-lag]))
            variance_contribution = (covariance / (std_r ** 2)) * daily_sr**2
            hac_adjustment += 2.0 * weight * variance_contribution

        variance_adjusted_total = max(variance_adjusted_core + hac_adjustment, 1e-
6)
        se_adjusted = np.sqrt(variance_adjusted_total / T)

        # Annualize our standard error metrics using the delta method
        # Var(c * X) = c^2 * Var(X) -> SE(ann_SR) = sqrt(factor) * SE(daily_SR)
        se_ann_adjusted = se_adjusted * np.sqrt(self._factor)

        # Calculate confidence intervals using critical values from standard
normal distribution
        z_crit = float(stats.norm.ppf(1.0 - self._alpha_level / 2.0))
        ci_lower = ann_sr - z_crit * se_ann_adjusted
        ci_upper = ann_sr + z_crit * se_ann_adjusted

        # Hypothesis Test H0: SR = 0 vs H1: SR != 0
        t_stat = daily_sr / np.sqrt(1.0 / T)
        p_val = float(2.0 * (1.0 - stats.norm.cdf(abs(t_stat))))

        return SharpeInferenceMetrics(
            daily_sharpe=daily_sr,
            annualized_sharpe=ann_sr,
            standard_error_iid_normal=se_iid_normal * np.sqrt(self._factor),
            standard_error_adjusted=se_ann_adjusted,
            confidence_interval_lower=ci_lower,
            confidence_interval_upper=ci_upper,
            p_value_zero_null=p_val
        )


# ---------------------------------------------------------------------------
# Validation Entrypoint
# ---------------------------------------------------------------------------
```

```
if __name__ == "__main__":
    np.random.seed(42)
    sample_size = 252

    # Simulate a typical equity CRB strategy return stream:
    # High kurtosis (fat tails) and mild positive autocorrelation
    innovations = np.random.standard_t(df=4, size=sample_size) * 0.01  # Fat-
tailed Student's t
    simulated_returns = np.zeros(sample_size)

    # Inject positive autocorrelation (AR(1) coefficient = 0.15)
    simulated_returns[0] = innovations[0]
    for t in range(1, sample_size):
        simulated_returns[t] = 0.0008 + 0.15 * simulated_returns[t-1] +
innovations[t]

    # Execute the inference calculations
    engine = CRBSharpeInferenceEngine(annualized_factor=252)
    metrics = engine.analyze_returns(simulated_returns, lag_hac=3)

    print("\n" + "="*65)
    print("  NOMURA RISK MANAGEMENT — STRATEGY PERFORMANCE INFERENCE  ")
    print("="*65)
    print(f"Point Annualized Sharpe Ratio    : {metrics.annualized_sharpe:.4f}")
    print(f"Standard Error (IID Normal)      :
{metrics.standard_error_iid_normal:.4f}")
    print(f"Standard Error (HAC Adjusted)    :
{metrics.standard_error_adjusted:.4f}")
    print(f"95% Adjusted Confidence Interval :
({metrics.confidence_interval_lower:.4f},
{metrics.confidence_interval_upper:.4f})")
    print(f"H0: SR=0 Asymptotic p-value       : {metrics.p_value_zero_null:.4e}")
    print("="*65 + "\n")
```

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4. Key Follow-Ups & Diagnostic Questions

Question 1: If daily returns exhibit an autoregressive profile $r_t = \rho r_{t-1} + \epsilon_t$, how does this modify the long-term standard error?

Answer: When returns exhibit an AR(1) serial dependence structure, the standard error of the sample variance changes significantly. Positive serial correlation ($\rho > 0$) causes variance estimators to underestimate the true long-term variability of the returns stream.

Under the Newey-West framework, the asymptotic variance of the sample mean scales by a variance inflation factor:

$$V_{\text{HAC}} = \sigma^2 \left(1 + 2\sum_{k=1}^{\infty} \rho^k\right) = \sigma^2 \left(\frac{1+\rho}{1-\rho}\right)$$

This inflation factor propagates through the Delta Method derivation. It expands the standard error of the Sharpe ratio, widening the calculated confidence intervals and reducing the statistical power of your performance validation tests.

Question 2: Why is the statistical power of a Sharpe ratio test often very low when evaluated over a 1-year horizon?

Answer: The statistical power function depends directly on the sample size T and the ratio of the true parameter value to its standard error. For an annualized Sharpe ratio of 1.5, the underlying daily Sharpe ratio is small: $\widehat{SR}_{\text{daily}} = \frac{1.5}{\sqrt{252}} \approx 0.0945$.

Because the daily point estimate is small relative to the baseline daily standard error ($\approx \frac{1}{\sqrt{252}} = 0.0629$), the sampling distribution under the alternative hypothesis overlaps significantly with the null distribution over a short 252-day horizon.

This overlap means that a high-Sharpe backtest can easily be generated by random market noise over a 1-year window. To achieve a reliable statistical power level ($1 - \beta \geq 0.80$) and rule out random luck, the strategy must be evaluated over a much longer historical backtest window.